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Rotational motion of the knee*

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ABSTRACT

This study deals with the quantitation of axial rotation at the knee. Passive rotation of the knee was measured at various degrees of flexion on 17 subjects with normal knees and 19 patients with unilateral anterolateral rotatory instability.

Normal subjects were found to have bilateral symmetric rotational knee motion at each angle of flexion tested. When the knee is in a position of flexion between 30 and 90°, there are approximately 45° external and 25° internal rotation. Rotatory motion decreases with further extension and, at 5° of flexion, the knee has 23° external and 10° internal rotation.

The knee with a torn anterior cruciate ligament and anterolateral rotatory instability was found to have an arc of rotation similar to the contralateral knee and to the control knees between 30 and 90° flexion. At 15° of flexion, a slightly greater arc of rotation was measured compared to normal knees. At 5° of flexion, a significantly greater range of external (to 41°) as well as internal (to 14°) rotation was measured.

Axial rotation is an important component of knee motion[4,18]; however, the normal range of rotation of the knee is not commonly known, and rotational motion is rarely quantitated in patients.

Excessive rotational force to the knee is known to tear ligamentous structures resulting in rotatory instability.[11,16,23] To our knowledge, however, abnormal increase in arc of rotation accompanying the instability has never been documented or clinically measured in patients with rotational instability.

The purpose of this article is to present a clinical method for and the results of measuring passive rotational motion of the knee. The aims of this study were to: (1) develop a clinical method of measuring rotational motion at the knee; (2) measure the range of rotational motion of the knee at different positions of flexion; (3) determine if excessive rotation is present in knees with a known rotatory instability; and (4) establish at which positions of knee flexion excessive rotation is maximal, so that this joint position could be used for clinical testing.

Results of clinical studies which have measured the ability of the tibia to rotate on the femur have shown a great deal of variation (Table 1).[6,12,17,19-21] One cannot directly compare values obtained by different investigators since testing methods have varied greatly and have not been described in some publications. Less rotatory motion at the knee has been measured in cadaver studies compared to clinical studies (Tables 1 and 2). Brantigan and Voschell[3] found that the range of external rotation was greater than the range of internal rotation. The greatest degree of external rotation occurred between 30 and 90° of flexion. Hallen and Lindahl[7] found the greatest degree of rotation to occur at 60° knee flexion.

The contribution of the anterior cruciate ligament to rotatory stability has been tested by division of that ligament in cadavers. Brantigan and Voschell[3] found that ligament division alone did not produce abnormal rotatory motion of the knee. After division, DePalma[4] found no abnormal rotatory motion in the extended position, but found that internal rotation "increased greatly" at 90° flexion. Abbott[1] stated that division of the ligament results in a slight increase of external rotation of the tibia, but did not mention at which angle of the knee flexion. Markolf et al.[14] found that sectioning the ligament produced only minor increases in full extension and produced no change when the knee was in flexion.

Thus, little information exists on how much rotation takes place in the normal knee, along with controversy regarding how this rotation is altered by loss of the anterior cruciate ligament.

MATERIALS AND METHODS

Passive rotational motion of knees was tested on 17 normal subjects and 19 patients with documented unilateral anterior

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cruciate ligament tears with chronic anterolateral rotatory instability. Normal subjects (eight males and nine females) were healthy volunteers who ranged in age from 17 to 28 years. None had any prior history of knee malfunction. Patients with anterior cruciate ligament tears (12 males and 7 females) ranged in age from 16 to 29 years with one exception who was 40 years old.

All patients met the following criteria:

History

•Twisting injury more than 6 months prior to testing.

•Onset of severe swelling within 2 hours of injury.

•Subsequent recurrent symptoms of giving way of the knee with pivoting motions.

Physical examination

•Positive anterior drawer sign at 20° flexion (Lachman test).

•Positive pivot shift phenomenon (anterior subluxation of lateral tibial plateau).

•No other knee instability elicited by clinical testing.

A torn anterior cruciate ligament was confirmed by arthroscopy or arthrotomy in all patients. In addition, the medial meniscus previously had been removed in nine patients, lateral meniscus in four, and both menisci in seven.

Rotatory motion of the tibia in relation to the femur was measured using a position potentiometer mounted on the input shaft of a Cybex II isokinetic dynamonmeter (Lumex, Bay Shore, NY) and was recorded on a strip chart recorder.

Subjects were tested in the side lying position (Fig. 1). The upper leg was crossed over to get it out of the way and to help stabilize the pelvis. Pelvis and thigh were stabilized by manual pressure. Leg, ankle, and foot were secured in a short leg orthosis to prevent ankle, subtalar, and foot motion. The foot was placed in an Elgin boot attached to the dynamometer input shaft, which was aligned with the longitudinal axis of the leg. The knee was positioned at angles of 90, 60, 30, 15, and 5° flexion. Full extension of the knee was recorded at 0° flexion instead of 180°[10] as in many earlier studies. In each position, passive internal and external rotation of the tibia in relation to the femur was measured. The neutral rotation position of "zero" degree rotation was estimated in each instance by aligning the tibial tubercle and lower leg with the femoral condyles. Patients were instructed to relax their muscles to allow the examiner to move the leg. Three warm-up trials were performed to assure relaxation. Then, three separate measurements of internal and external rotation were taken. Thirteen subjects were retested by the same examiner at an interval of 1 year.

Data analysis was based on the mean of three consecutive trials of internal and external rotation on each subject at each joint angle.

TABLE 1
Clinical studies—degrees of rotation at knee

Angle of knee flexion (deg)	30–45°	60°	90°	120–135°
Meyer (7)	32	33	42	52
Fick (8)			50	70
Quain (9)		50		
Ross (10)	41		37	39
Lang et al. (11)			52	
Ruetsch et al. (12)			36	
Osternig et al. (13)	50		58	
Zarins et al.	72	73	74	

RESULTS

Normal knees (Fig. 2)

•No significant difference was found between the right and left knees in the range of passive rotational motion at each position of flexion tested.

•No significant change in the arc of rotational motion was found in measuring at 30, 60, or 90° of knee flexion.

•As the knee was extended from 15 to 5° of flexion, rotatory motion decreased. At 15° of flexion, the knee had 47° external rotation and 16° internal rotation, a total range of 33° rotation.

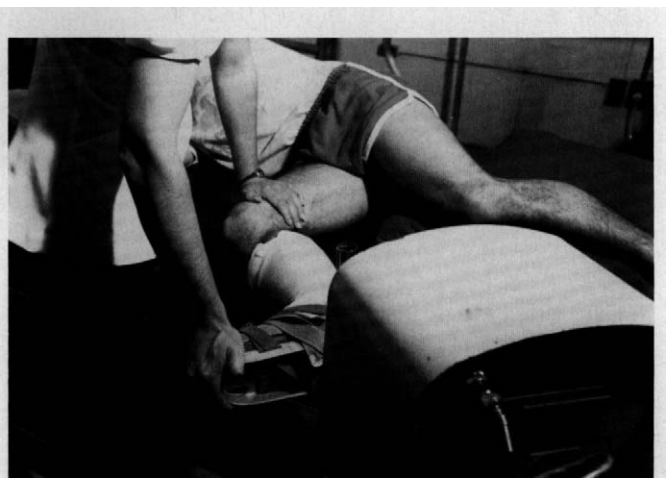

Figure 1. Method used to measure rotational motion of the knee in sidelying position (see text).

TABLE 2
Cadaver studies—degrees rotation at knee

Angle of knee flexion (deg)		0°	20–30°	60°	90°	Full flexion
Brantigan et al. (14)	1941	0			6–24	4–21
DePalma (2)	1954	0	30–38		Same as 30° flexion	
Hallen et al. (15)	1965	11	25	29	26	
Markolf et al. (17)	1976		10		24	26

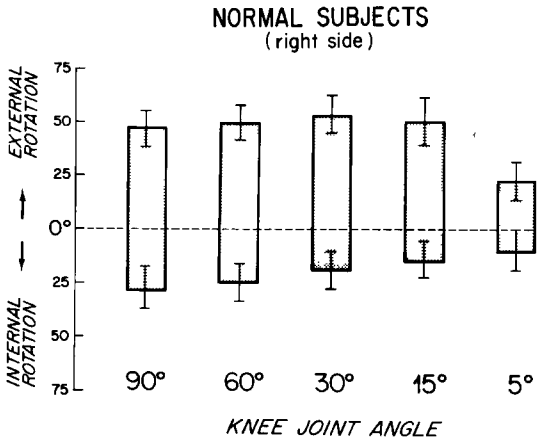

Figure 2. Results of measuring rotational motion of normal knees at varying degree of knee flexion.

Knees with torn anterior cruciate ligament (ACL) and anterolateral rotatory instability (Fig. 3)

•At 30, 60, and 90° of flexion, the range of rotation of the unstable knee was similar to the patient's normal knee and to other normal knees tested.

•At 15° of flexion, the knee with a torn anterior cruciate ligament had a slightly greater arc of internal and, therefore, total rotation.

•At 5° of flexion, the knee with a torn anterior cruciate ligament had significantly more in external and internal rotation compared to the normal knee as well as the control knees ($P \leq 0.05$).

Test-retest reliability of method

Normal knees of 13 subjects were retested by the same examiner at an interval of 1 year (Table 3).

DISCUSSION

We measured range of rotation in the sidelying position as we believe that making the measurement in this position has several advantages. The thigh and pelvis are easily stabilized by downward pressure against the table and the effect of gravity is eliminated. If the leg is allowed to hang free, such as when the patient sits at the edge of the table, gravity distracts the joint. If the foot is resting on the measuring device, the joint is stabilized by axial loading. If the subject is tense, less rotation is measured. It is easier to attain patient relaxation in the side-lying position.

There were *difficulties* in testing rotation with this method, however. The examiner must be skilled in testing method. It is difficult to find the "zero position" of rotation (neutral). Thighs of subjects who are obese are difficult to stabilize, especially as the knee comes into extension. This stabilization is hard to attain especially in patients with tibial torsion, genu valgum, genu varum, and other malalignment syndromes.

To measure our testing method reliability, the same examiner retested rotation of the knee on the same subject 1 year later. The results (Table 3) show that accuracy of measuring internal and external rotation is not as high as measurement of total rotation, which is reliable. The lower

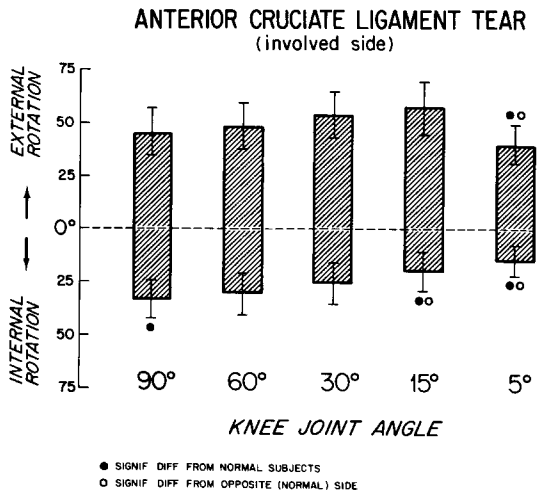

Figure 3. Results of measuring rotational motion of knees with a torn anterior cruciate ligament.

TABLE 3
Test-retest reliability correlation coefficients of 13 subjects tested by the same examiner

Angle of knee flexion	IR	ER	Total
90	0.58	1.00	0.96
60	0.59	0.95	0.93
30	0.57	0.99	0.93
15	0.80	0.94	0.92
5	0.34	0.84	0.95

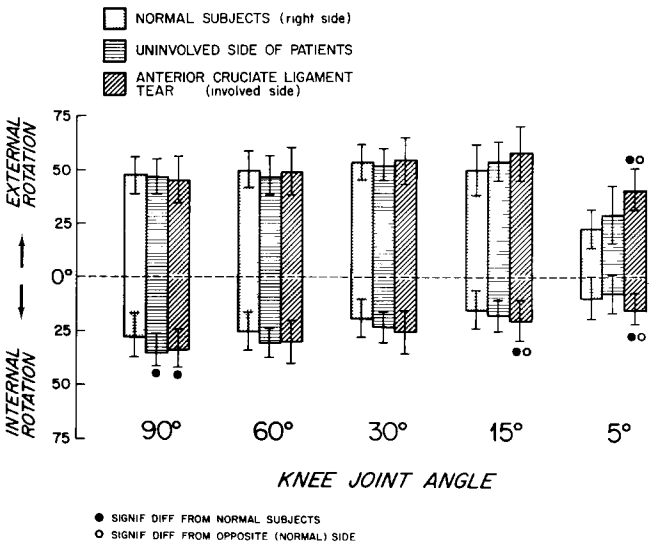

Figure 4. Comparison of range of rotation of normal knees and knees with torn anterior cruciate ligament.

correlation coefficient for internal rotation points up the difficulty of defining zero rotation. In the measurement of internal rotation, the foot and leg are moved against gravity which may explain the difficulty in the assessment of the end point of rotation.

Our results differ from previously reported studies on axial rotation of the knee in that we measured a greater degree of internal, external, and, therefore, total rotation. Nicholas[16] has clinically noted similar findings: a large arc of rotation of the knee in quick changes of direction while running.

There are several possible explanations for the wide range of motion found in this study:

•Testing in the sidelying position relaxes the subject and eliminates muscular forces, effects of gravity, and axial loading. Wang and Walker[25] showed that an axial load of 983 newtons on the knee joint at 25° reduced its rotatory laxity by 80%. Markolf et al.[13] measured laxity decreases of 13 to 25% in this joint position with 925 newtons of load.

•Undetected motion at the hip could give falsely high rotational values measured at the knee. If this were the case, however, one would expect to have found a greater range of rotation as the knee came into extension. Since this situation did not occur, we believe we were able to stabilize the thigh adequately and eliminate hip motion. A great deal of pressure can be easily applied to the thigh in the side-lying position. Crossing the opposite leg also helps stabilize the pelvis.

Although it is possible that motion in the foot or ankle was not completely eliminated in this study, we believe we were able to stabilize these joints adequately with the splint. The subjects were not able to perceive motion in these joints during the test.

In accordance with earlier studies[3,25] the range of total rotation did not change significantly between 30 and 90° of flexion. It was not until the knee extended to 15° of flexion and then to 5° of flexion that rotation decreased. Rotation at the knee was not tested at 0° flexion because we were unable to stabilize the thigh and prevent hip rotation in this knee position. The symmetry between the right and left knees in normal subjects is also in accordance with previous studies.

In this study, no abnormal rotatory motion of the knee was found between 30 and 90° in subjects with a torn anterior cruciate ligament and anterolateral rotatory instability. Hallen et al.[7] found 60° to be the position of excessive rotatory motion in the cadaver knee with severed medial collateral and anterior cruciate ligaments. Hughston[7] stated that ligamentous stability is more important in the flexed position than in the standing position of extension. According to Hughston, most activities in sports occur in a position of knee flexion between 45 and 90°. We could not document excessive rotation at the knee in positions of flexion past 30° in patients with anterolateral rotatory instability.

The most significant finding in this study is the measurement of statistically significant increase in rotation of the knee with anterolateral rotatory instability at 5° of knee flexion. This finding supports the concept that the anterior drawer sign is a more specific test for anterior cruciate ligament function in relative extension (Lachman test) than at 90° of flexion. The finding also correlates with the pivot-shift phenomenon (anterior subluxation of the lateral tibial plateau), which we believe is the most important physical finding in the knee with instability due to a torn anterior cruciate ligament.

An abnormal increase in the range of rotation in slight flexion of the knee also fits with observations on the mechanism of injury to the anterior cruciate ligament. Wang et al.[24] described hyperextension in internal rotation of the tibia to be the mechanism of isolated tear of the anterior cruciate ligament. Our own clinical observation is that most tears of the anterior cruciate occur from excess force from deceleration and twisting (usually external rotation of tibia) in a position of slight knee flexion.

All patients with anterolateral rotatory instability in this study had had a meniscectomy and seven had both menisci removed. It has been our clinical observation that removal of a meniscus may make an instability more severe. Seale et al.[23] showed in cadaver experiments that medial meniscectomy increases both internal and external rotation and that severing the anterior cruciate ligament after dual meniscectomy increases rotation especially at 20° of flexion. Relative roles which ligaments and menisci play in controlling rotation needs further study.

The findings of abnormal increase in external and internal rotation of the knee at 5° of flexion and normal rotational motion in greater degrees of knee flexion should make us reexamine the technique of positioning the tibia in extreme external rotation in extraarticular reconstruction for anterolateral rotatory instability.[2,5,11] If the injury to the anterior cruciate ligament has occurred from excessive external rotation, and if the unstable knee has excessive external rotation as shown in this study, the use of extreme rotation position could be open to question.

The greatest source of error in the method used in this study seems to be the determination of the neutral or "zero" point of rotation. While the accuracy of determining the neutral position needs further study, we believe that the total range of rotation can be accurately measured using this method.

The method described in this paper for measuring knee rotation is time consuming. Improved ways of clinically measuring rotation should be looked for to make it easier to measure rotation in a clinical setting. The data presented in this paper should be viewed as pilot information pointing out the need for measuring knee rotation. Suggestions for further study include use of surface electrodes to be sure the muscles are relaxed, the development of a more accurate method of determining the zero point of rotation, and a method of determining the amount of forward tibial translation which might be taking place during rotation.

CONCLUSION

•Total range of rotation of the knee can be accurately measured.

•Normal subjects have bilateral symmetric rotational motion at each angle of flexion tested.

•The range of rotational motion does not change significantly between 30 and 90° of flexion in normal subjects. Abnormal increase in rotation is not found in patients with anterolateral rotatory instability at these degrees of flexion.

•The knee with anterolateral rotatory instability postmeniscectomy has an abnormal increase in the range of rotation between 5 and 15° of knee flexion. This appears to be the best position for measuring tibial rotation. Testing rotation at degrees of flexion greater than 15° yields less significant information than testing done in relative extension.

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DISCUSSION

Donald B. Slocum, MD, I would like to congratulate Dr. Zarins and his associates on a most interesting study. His method appears to be a distinct advance over a previous period analysis in determining the rotation of the vertical axis of the tibia. His machine firmly fixes the foot and ankle to the tibia; the lateral position provides fixation of the pelvis as well as the elimination of gravity and muscle tension. When applied to the problem of anterior cruciate absence or insufficiency in combination with anterior lateral rotatory instability, it should be somewhat refined in the skin-bone relationship and in the human factor. Since the Cybex records only 1° of freedom of motion, it primarily records 1° of motion about the vertical axis of the tibia. Since all joints have 6° of freedom of motion, the translatory motion of the knee which follows anteriorly due to the lack of the anterior cruciate constraint immediately due to anterior lateral rotatory instability had led to change in the vertical axis of rotation to a more anterior medial position through these translatory factors. It is my hope that Dr. Zarins will continue with his research to solve the problem of translation of what we call anterior lateral rotatory instability.

Authors' Reply: Dr. Slocum, we regard this as a pilot study and plan to pursue the quantitation of rotational motion of the normal and unstable knee to try to answer some of these questions. Thank you.